Fiscal Year:	FY 2026	**Task Last Updated:**	FY 02/10/2026
PI Name:	Strangman, Gary E Ph.D.		
Project Title:	Personalized Performance Optimization Platform (P-POP)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	strang@nmr.mgh.harvard.edu	**Fax:**	FY
PI Organization Type:	NON-PROFIT	**Phone:**	617-724-0662
Organization Name:	Massachusetts General Hospital		
PI Address 1:	Department of Psychiatry		
PI Address 2:	149 13th Street, Suite 2651		
PI Web Page:			
City:	Charlestown	**State:**	MA
Zip Code:	02129-2020	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019-2020 HERO 80JSC019N0001-HHCBPSR, OMNIBUS2: Human Health Countermeasures, Behavioral Performance, and Space Radiation-Appendix C; Omnibus2-Appendix D
Start Date:	02/23/2021	**End Date:**	09/30/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2025 per A. Beitman, NASA-JSC (Ed., 8/14/25). NOTE: End date changed to 02/22/2026 per S. Mack-Phillips, NASA-JSC (Ed., 1/20/25).		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Ivkovic, Vladimir Ph.D. (Massachusetts General Hospital) Stankovic, Aleksandra Ph.D. (Massachusetts General Hospital) Zhang, Quan Ph.D. (Massachusetts General Hospital) Maes, Patricia Ph.D. (Massachusetts Institute of Technology) Kosmyna, Nataliya (Massachusetts Institute of Technology)		

Grant/Contract No.:	80NSSC21K0669
Performance Goal No.:	
Performance Goal Text:	
Task Description:	BACKGROUND: The environmental conditions of prolonged spaceflight pose significant psychological risks for astronauts. In particular, long duration exposure to an isolated and confined environment plus circadian disruption and numerous other factors can contribute to adverse cognitive or behavioral events and compromise mission safety and/or mission success. In order to mitigate against mission-related disruptions arising from decrements in behavioral health and performance, NASA needs a set of validated strategies on-board to both maintain and/or restore psychological well-being and operational effectiveness. This project sought to refine and empirically assess a platform technology designed to monitor and guide crewmembers towards optimal physiological and mental states for current or future tasks via personalized manipulation of the individual's personal work environment. Our personalized, closed-loop, brain-computer interface (BCI)-based intervention approach not only helps maintain individual behavioral functioning but, by supporting individual behavioral health, can help improve team performance as well. Our four specific aims were as follows: AIM 1: Perform a detailed risk assessment of factors that contribute to personal (and team) dysfunction, particularly in isolated, confined, and extreme environments. AIM 2: Develop a personalized performance-optimization platform (P-POP) based on closed-loop/feedback that integrates physiological sensing with augmentation of the astronaut's local working environment (e.g., audio, haptics, light). AIM 3: Characterize the ability of P-POP to improve key performance capabilities including attention, response time, memory, cognitive control, and operationally-relevant performance. AIM 4: Assess the feasibility, acceptability, and efficacy of our proposed platform for use in individuals and teams via empirical testing during long-duration spaceflight analogs. HYPOTHESES: (Hyp1) The novel P-POP will provide real-time physiological monitoring to enable the personalized manipulation of the local work environment—both in the lab and in Human Exploration Research Analog (HERA). (Hyp2) Our targeted work environment modulations (e.g., sound, haptics, light) will generate significant improvements in individuals' cognitive and operational performance. DELIVERABLES: Our project sought to generate the following deliverables: (1) a review of factors that contribute to poor individual and team performance in spaceflight and isolated, confined, and extreme (ICE) settings as well as the efficacy of simple environmental manipulations to help mitigate behavioral health risks; (2) a novel BCI-based platform technology capable of real-time tracking of psychological and behavioral health markers and providing targeted augmentation of the local work environment to manipulate those markers; (3) an evaluation of the feasibility, acceptability, and efficacy of the proposed platform technology in multiple settings including testing in a spaceflight analog. SIGNIFICANCE: This work provides NASA with a novel and scalable platform technology for on-board behavioral health management—adapting the local working environment via biosensing and feedback. The approach is personalized and closed-loop, automatically guiding individuals away from less-optimal states (as assessed by physiological measurements) towards more-optimal states, with minimal effect on the environment that might affect other crew members. We expect the approach to help maintain and improve individual performance and, as a consequence, also improve team performance. The P-POP system does not require video displays or graphics, as per the solicitation. Importantly, however, the platform was designed for future augmentation via other countermeasure approaches (e.g., visual, olfactory), depending on the needs and capabilities of any particular exploration mission. On Earth, such a platform could have considerable utility for optimizing human performance in a wide range of workplaces.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The goal of P-POP was to develop a personalized/wearable system that can help optimize and enhance user cognitive and spaceflight operations performance. The system incorporated relatively low-tech "countermeasures"--haptic stimuli, auditory stimuli and lighting stimuli--to help optimize the user's alertness, attention and arousal levels depending on the user's current state and the requirements of the task. The optimization involved a "closed loop" concept whereby physiological sensing helps identify the user's present state, then optimization via deployment of zero or more countermeasures (CMs), with physiological sensing helping also determine the consequences of the CMs, thereby allowing for real-time, adaptive feedback-based optimization. All this was done "on platform" (edge computing), not requiring any cloud services at all. Such an approach--being based on feedback from the individual--is inherently personalized: each individual crewmember, by using their own system, can achieve different simulation/relaxation goals simultaneously, even in the same physical workspace. The deployment of CMs in the users "local" environment (e.g., via earbuds or worn tactors or eyeglasses) further supports the personalization—by minimizing the disruption to other crew members. On Earth, such a platform could have similar utility for optimizing human performance in a wide range of office or remote workplace settings.
	The objective of this project was to develop a personalized performance optimization platform, which functions by sensing physiological parameters from the user, specifically including brain activity via electroencephalography (EEG), interpreting that data to identify the user's state (e.g., under- vs. over-stressed, inappropriate arousal level, inattention, fatigue, etc.), and then delivering one or more countermeasures via haptic, auditory, or lighting-based stimulation to optimize that state and thereby maintain or enhance cognitive/behavioral performance. The goal was to accomplish this in a continuous, real-time fashion for continuous performance optimization in operational settings. P-POP development was conducted in phases. Phase 1 involved the development and integration of the auditory CM into the base/initial AttentivU hardware platform, specifically using music to help optimize behavioral performance. This development/integration effort was immediately followed by tests of the auditory CM in n=24 volunteers over a 6-week, self-deployed (at home) testing protocol involving cognitive testing. This work revealed an ability to positively affect performance by varying auditory (music) stimuli. In parallel with this testing, we initiated Phase 2, which incorporated

Task Progress:	the haptics CM into the P-POP platform. This process involved the development and linking of a multi-tactor somatosensory sleeve or bracelet that could serve to increase or decrease arousal and attention levels based on the current brain state. Again, this development and integration work was followed immediately by tests of that CM in n=24 volunteers in another 6-week, self-deployed (at home) testing protocol, in this case involving both cognitive testing and operationally-relevant testing with our enhanced SpaceDock2 spacecraft docking simulation. In parallel with Phase 2 testing, we also started Phase 3 development to add a light-based CM to the P-POP platform. This CM was designed to leverage recent research showing that 40Hz flickering light can enhance attention and alertness, whereas 10Hz light can be used to induce calmer mental states. As with music and haptics, this makes a third CM that can help modulate the user's psychophysiological state and “nudge” them towards the state of peak performance. Light-enabled P-POP prototypes were generated, along with the development of next-generation cognitive testing (CogScope) and operationally-relevant assessment (SpaceDock2). Testing of the Phase 3 system could not be completed due to the premature termination of the award. Phase 4 (Aim 4) of the project was scheduled to involve deploying the three-CM P-POP platform in the NASA Johnson Space Center (JSC) Human Exploration Research Analog (HERA) facility during Campaign 8 as a CM to isolation in confinement during a HERA analog campaign. That work was also not conducted due to the premature termination of the P-POP award due to Administration budget constraints.
Bibliography Type:	Description: (Last Updated: 07/08/2026)
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Abstracts for Journals and Proceedings	Kosmyna N, Seligson A, Cheatham K, John J, Stankovic A, White B, Ivkovic V, Maes P, Strangman G. "P-POP: Deploying music and auditory stimulation for real-time improvement of performance, relaxation and well-being using physiological sensing." 2022 NASA Human Research Program Investigators’ Workshop, Virtual, February 7-10, 2022. Abstracts. 2022 NASA Human Research Program Investigators’ Workshop, Virtual, February 7-10, 2022. , Feb-2022
Abstracts for Journals and Proceedings	Stankovic A, Kosmyna N, White B, Ivkovic V, Maes P, Strangman G. "P-POP: A personalized performance optimization platform for long duration spaceflight." 2022 NASA Human Research Program Investigators’ Workshop, Virtual, February 7-10, 2022. Abstracts. 2022 NASA Human Research Program Investigators’ Workshop, Virtual, February 7-10, 2022. , Feb-2022
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